



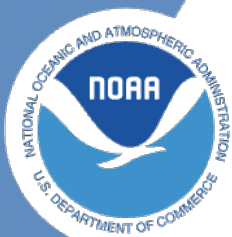
NOAA Technical Memorandum NMFS-XXX-##

Makara Data Submission Guide

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Genevieve Davis

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National Oceanic and Atmospheric
Administration
National Marine Fisheries Service
Northwest Fisheries Science Center



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Makara Data Submission Guide

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Executive Summary

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1 General_Information

1.1 General Information

This guide provides data contributors with instructions, field definitions, and examples for how to enter acoustic data and metadata into the template tables and submit them for archiving in the Northeast Fisheries Science Center's (NEFSC) Makara Database. Data that are archived in Makara may also be displayed on the Passive Acoustic Cetacean Map (PACM), with permission from the data owner(s).

2 2_Document_History_and_Revisions

2.1 Document History and Revisions

Ver- sion	Date	Author	Description
1.0	May 1, 2025	Juan Lianne Wilder	Document Created
1.1	June 10, 2025	Jefrey Walker	Updated "Data Table Template Instructions" to clarify that filenames and folder structure must match "templates" sub-directory.
1.2	August 26, 2025	Jefrey Walker	- Renamed the organization_code to deployment_organization_code in the analyses and detections tables to clarify that this field refers to the organization that owns the deployment. Also added analysis_organization_code to the detections table. Both tables now include both the deployment_organization_code (organization that the deployment and recording(s) belong to), and analysis_organization_code (organization that the analysis belongs to). - Added track_uri to tracks table for linking tracks to external files in cloud storage (useful for linking track metadata to high resolution data not stored in makara) - Fixed minor typos

2 2_Document_History_and_Revisions

1.3	October 7, 2025	Jefrey Walker	• Clarified that deployments, recordings, analyses and detections are the primary data tables, but not always required for each submission • Added Sensor Datasets and Sensor Values templates • Added deployment_realtime and deployment_url fields • Updated JSON guidelines and examples • Set default value for recording_n_channels to 1 • Fixed analysis_processing_types to have values of POST_PROCESSED and REAL_TIME (underscores instead of dashes) • Clarified descriptions of analysis_release_data and analysis_release_pacm
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3 Contact_Information_for_the_Makara_- Database_Support_Team

3.1 Contact Information for the Makara Database Support Team

If you have questions or issues regarding your data submission, please contact nmfs.nec.pam-data@noaa.gov.

4 4_Tips_and_Recommendations

4.1 Tips and Recommendations

- Timezones/Timestamps: All timestamps that are entered into the data submission templates should include the timezone (even if they are in UTC; e.g., 2023-10-12T18:09:05+0000). The NEFSC generally recommends that all recordings are made in UTC. Other timezones are okay, so long as the timezone information is included within the timestamp (e.g., 2023-10-12T18:09:05+0500 to indicate that the audio was recorded in Eastern Standard Time).
- Scenarios and Examples: It is highly recommended that data contributors look through our Scenarios section and the 'examples' template folder in the downloadable zipped folder on the Passive Acoustic Reporting System Templates webpage prior to entering their data into the data submission templates. There may be an example that is similar to their situation.

5 5_Data_Table_Template_Instructions

5.1 Data Table Template Instructions

To submit data and metadata to the Makara database, the information must be organized into template tables. The data template tables and example tables are provided in a downloadable zipped folder on the Passive Acoustic Reporting System Templates webpage, and are described in more detail below. Some of the tables are required, but do not necessarily need to be submitted at the same time. Other tables are optional, but it is highly recommended that the data contributor fills out any tables that are applicable to their deployment(s). Within each table, there are required and optional (but recommended if applicable) fields. The required tables/fields are the bare minimum of what metadata should be submitted.

Within the template zipped folder is a sub-directory named "templates/". This sub-directory contains empty CSV files that can be used as starting templates for filling out each table. Each data submission should contain one or more of these CSV files with data rows properly entered based on the formats described below. The names of each CSV file as well as the folder structure must match what is in the templates folder. Data cannot be submitted as an Excel Workbook (*.xlsx file).

6 6_Primary_Data_Tables

7 7_Optional_Data_Tables

8 8_Reference_Tables_Maintained_by_- NEFSC

9 9_Data_Type_Formatting_Instructions

10 10_Data_Table_Field_Definitions

11 11_Deployments_Table_Field_ Definitions

12 12_Recordings_Table_Field_Definitions

13 13_Analyses_Table_Field_Definitions

14 14_Detections_Table_Field_Definitions

15 15_Devices_Table_Field_Definitions

16 16_Projects_Table_Field_Definitions

17 17_Sites_Table_Field_Definitions

18 18_Recording_Intervals_Table_Field_- Definitions

19 19_Tracks_Table_Field_Definitions

20 20_Track_Positions_Table_Field_Definitions

21 21_Sensor_Dataset_Field_Definitions

22 22__Sensor__Values__Table__Field__- Definitions

23 23_Frequently_Asked_Questions

24 24_Example_Scenarios